REST - Representational State Transfer. RESTful API, is a simple, uniform interface that is used to make data, content, algorithms, media, and other digital resources available through web URLs.

Client initiates a request -> **Request travels over the network** -> **Server processes the request** -> **Performs Database operations** -> **Response generated -> Response returned.**

Your application makes an HTTP request to an API endpoint. The request moves through the internet over HTTP/HTTPS to reach the server. The API validates input and performs logic or functionality. If needed, the server queries or updates the database. The server returns a structured response (often in JSON) with a status code, headers, and a body. The client receives the response and processes it, such as rendering a UI, logging, or making further calls.

Benefits of REST

- **Scalability**: Statelessness and cache control make horizontal scaling straightforward.
- **Flexibility**: Multiple representations (usually JSON) and content negotiation fit many clients.
- **Decoupling**: Client–server separation enables independent iteration.
- **Lightweight**: Familiar HTTP semantics and smaller payloads suit mobile and IoT scenarios.

The most commonly used HTTP methods to perform CRUD operations in REST APIs

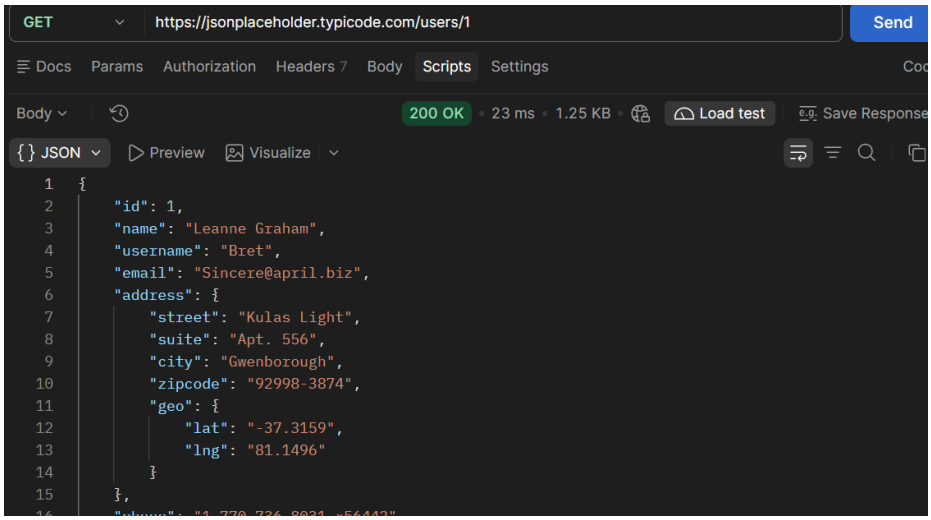
- GET - Reads data
- POST - Creates data
- PATCH - Updates a part of data
- DELETE - Deletes the data

Understanding the URL: Eg. https://jsonplaceholder.typicode.com/posts/1

- https:// - Communication Protocol
- jsonplaceholder.typicode.com - API server domain
- users - Resource
- /1 - Specific post required

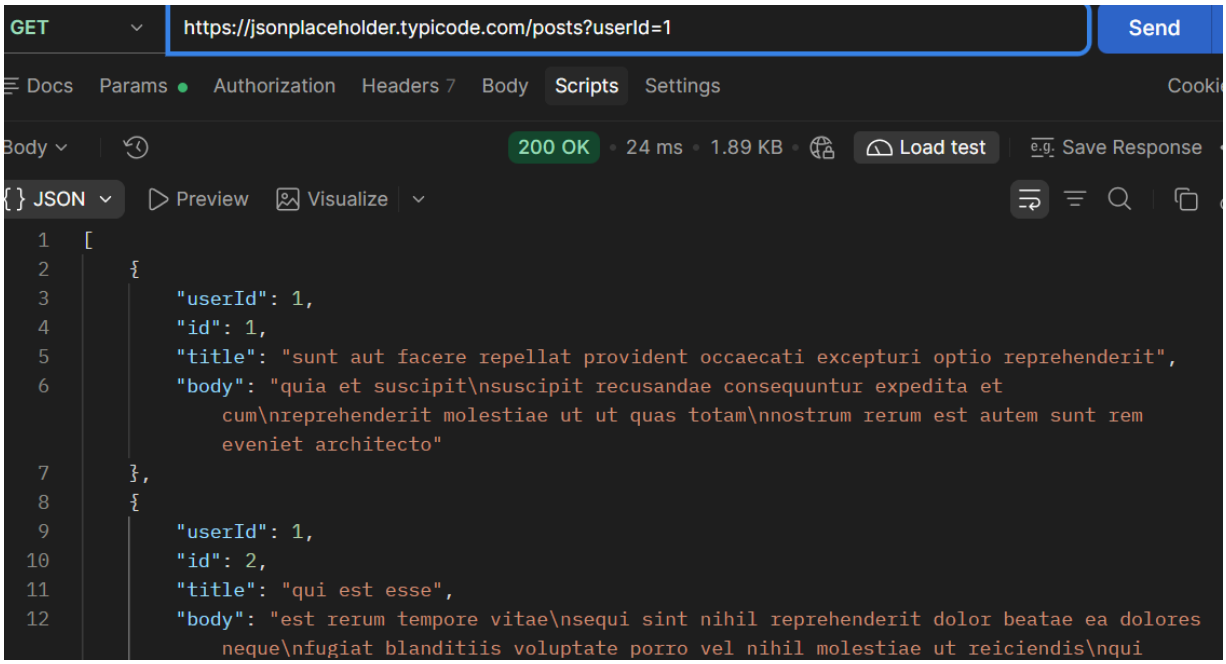
GET method:

Retrieves data of the user whose id is 1



Eg. https://jsonplaceholder.typicode.com/posts?userId=1 - Query parameter

Retrieves posts of those whose userId is 1

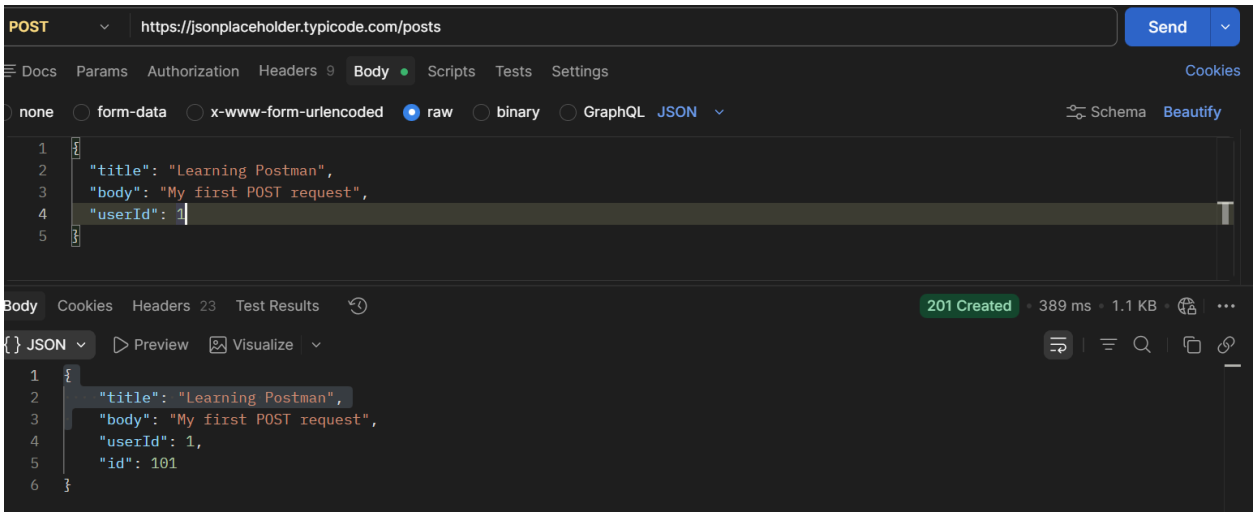


POST method:

Sending data using POST method

Eg. https://jsonplaceholder.typicode.com/posts

Add request to body: Body tab -> raw -> JSON



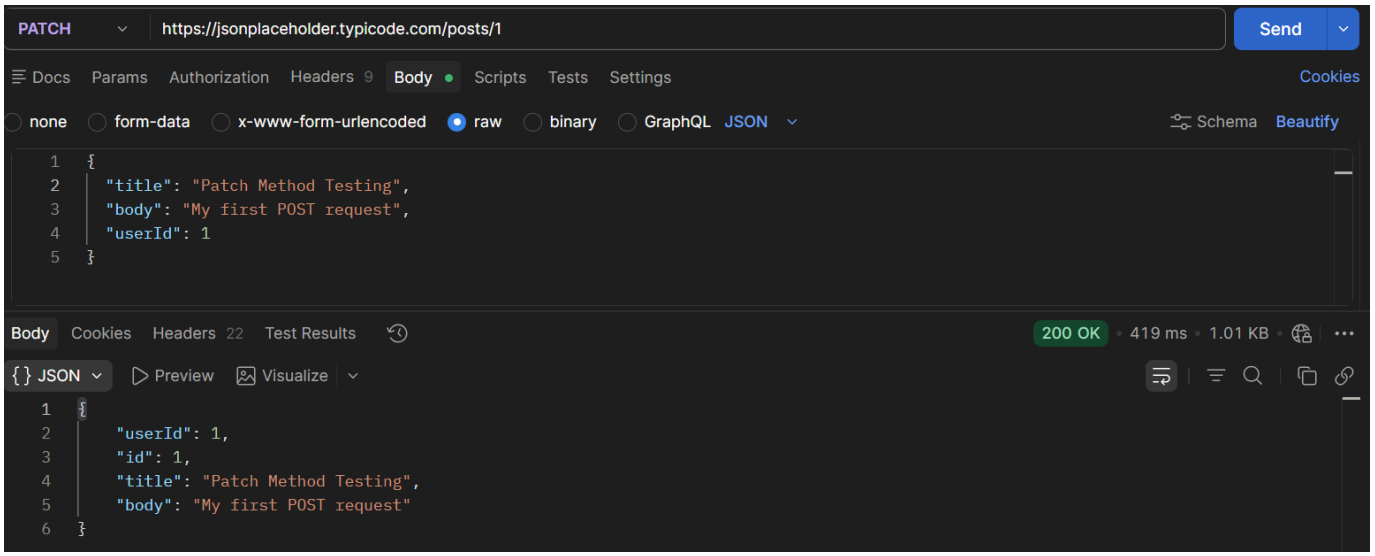
PATCH method

Update a post by selecting a url and changing the part of data that needs change.

Eg. http://jsonplaceholder.typicode.com/posts/1

Add request to body: Body tab -> raw -> JSON

Make the title to “Patch Method Testing”



DELETE method

Delete a post by selecting the url https://jsonplaceholder.typicode.com/posts/1 and leaving the json empty

